

Curriculum Vitae Umanga Gunasekera

Education

01/09/2017–10/25/2021

PhD (Veterinary Population Medicine), Minor in Epidemiology,
University of Minnesota, Minneapolis (United States)

Thesis title: Spatial and molecular epidemiological approaches for evaluating dynamics of foot-and-mouth disease virus in India and Vietnam.

Advisors: Kimberly VanderWaal

01/10/2013–30/09/2015

Master of Veterinary Public Health, Center for Asia Pacific, Freie University of Berlin and University of Chiangmai, Berlin, and Chiangmai

Thesis title: Risk factors regarding the Foot and Mouth Disease spread in the country and study the current FMD situation in Sri Lanka.

16/10/2006–14/02/2011

Bachelor of Veterinary Medicine, BVSc. (2nd Class-Upper division)

Faculty of Veterinary Medicine and Animal Sciences, University of Peradeniya, Kandy (Sri Lanka)

Professional Experience

Government Veterinary Surgeon 10/01/2015–07/01/2017, Department of Animal Production and Health (Sri Lanka)

1. Animal disease surveillance.
2. Supporting the livelihood of the community by sustaining animal health and animal production in the area.
3. Attended clinical cases (bovine, swine, and poultry) both at the hospital and by ambulatory visits. Conduct field visits regarding Poultry farms and Dairy farms.
4. Provided knowledge and advice to farmers to uplift their livelihood by avoiding health and zoonotic problems, selecting good animals for rearing purposes, and preventing the spreading of contagious diseases and zoonotic infectious diseases.

Diplomate in Veterinary Preventive Medicine, American College of Veterinary Preventive Medicine, 2023 to present

Registered as a veterinarian in the Veterinary Council of Sri Lanka (Registration number 1740)

Post-doctoral fellow, Center for Animal Health and Food Safety -- OIE Collaborating Center & FAO Reference Center, College of Veterinary Medicine, University of Minnesota

Supervisor: Anders Perez

6/12/2021- 30/6/2025

1. Developed and validated mathematical models for the transmission dynamics and early detection of critical transboundary and emerging infectious diseases (foot and mouth disease)
2. Utilized Bayesian spatio-temporal regression and network analysis to identify high-risk areas and transmission pathways in endemic settings, directly informing containment strategies.
3. Obtain and apply knowledge of new data analysis methods such as quantitative risk analysis and machine learning methods.
4. Initiate and conduct collaborative research between the university and the government sectors (Foreign Animal Disease Research Unit, USDA-ARS, Plum Island Animal Disease Center, and the Center and the University of Minnesota, Sri Lanka, Bangladesh)

5. Involvement in outreach activities related to Progress Vet. Progress Vet is an educational program conducted by the Center for Animal Health and Food Safety to educate practicing veterinarians in different countries in Africa and East Asia towards improving surveillance and epidemiological studies.

Graduate Research Assistant, College of Veterinary Medicine, University of Minnesota

08/17/2017-10/29/2021

1. Researched and published three manuscripts related to molecular epidemiology and spatial characteristics of the Foot and Mouth Disease (FMD) virus.
2. Designed and executed complex data pipelines combining pathogen genomics and epidemiological data using Bayesian phylogenetic analysis.
3. Analyzed large-scale, surveillance datasets in R to model infectious disease spread, improving forecasting accuracy for emerging outbreaks.
4. Presented quantitative findings and disease mitigation strategies to diverse stakeholders, bridging the gap between theoretical modeling and public health practice.

Publications

Submitted

1. **Gunasekera Umanga**, Puvanendiran Sumathy, Pradeep L. Kumarawadu, Arzt Jonathan, Perez Andres, Humphreys John, “Designing disease control zones: foot-and-mouth disease in Sri Lanka”. Submitted.

Published

2. **Gunasekera Umanga**, Nguyen Thi Diep, Pham Thanh Long, Vo Dinh Chuong, Arzt Jonathan, Perez Andres, “An overview of epidemiological tools that support the progress of foot and mouth disease control in Southeast Asia”. Pathogens. 2026 August. <https://www.mdpi.com/2076-0817/15/9/893>
3. **Gunasekera Umanga**, Fazi Mohamed A., Bartels Chris, Punyapornwithaya Veerasak, Perez Andres, VanderWaal Kimberly, Makau Dennis, 2024. “Optimizing risk-based disease control strategies through network analysis: Case insights from FMD in Sri Lanka”. Journal of Preventive Veterinary Medicine. 2025 October. <https://doi.org/10.1016/j.prevetmed.2025.106727>
4. **Gunasekera Umanga**, VanderWaal Kim, Jonathan Arzt, Andres Perez, “Foot and Mouth Disease reproduction number: a scoping review”. Frontiers in Veterinary Medicine. 2025 May. <https://www.frontiersin.org/journals/veterinaryscience/articles/10.3389/fvets.2025.1576974/full>
5. **Gunasekera Umanga**, Alkhamis Moh A., Puvanendiran Sumathy, Das Moumita, Kumarawadu Pradeep L, Sultana Munawar, Hossain Anwar, Arzt Jonathan, Perez Andres. 2024, “Ecological Niche Modeling for Surveillance of Foot-and-Mouth Disease in South Asia”. Plos One, April 2025. <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0320921>
6. **Gunasekera Umanga**, VanderWaal Kimberly, Arzt Jonathan, Perez Andres, “Methods to estimate the between-population level effective reproductive number for infectious disease epidemics: Foot-and-Mouth Disease (FMD) in Vietnam”. Transbound Emerg Dis. 2024 Nov 14; <https://onlinelibrary-wiley-com.ezp3.lib.umn.edu/doi/full/10.1155/2024/4114217>
7. **Gunasekara, Umanga**, Miranda R. Bertram, Nguyen Van Long, Phan Quang Minh, Vo Dinh Chuong, Andres Perez, Jonathan Arzt, and Kimberly VanderWaal. 2023. “Phylogeography as a Proxy for Population Connectivity for Spatial Modeling of Foot-and-Mouth Disease Outbreaks in Vietnam.” Viruses 15 (2): 388. <https://doi.org/10.3390/v15020388>
8. **Gunasekera, Umanga**. “Implications for a successful FMD control program in Sri Lanka”. Sri Lanka

Veterinary Journal, 2021, 68(1-2):17-29. [10.4038/slv.v68i1-2.65](https://doi.org/10.4038/slv.v68i1-2.65)

9. **Gunasekera, U.**; Bertram, M.R.; Dung, D.H.; Hoang, B.H.; Phuong, N.T.; Hung, V.V.; Long, N.V.; Minh, P.Q.; Vu, L.T.; Dong, P.V.; Perez, A.; VanderWaal, K.; Arzt, J. Use of Slaughterhouses as Sentinel Points for Genomic Surveillance of Foot-and-Mouth Disease Virus in Southern Vietnam. *Viruses* 2021, 13, 2203. <https://doi.org/10.3390/v13112203>
10. **Gunasekera U**, Biswal JK, Machado G, Ranjan R, Subramaniam S, Rout M, Mohapatra JK, Pattnaik B, Singh RP, Arzt J, Perez A, VanderWaal K. Impact of mass vaccination on the spatiotemporal dynamics of FMD outbreaks in India, 2008-2016. *Transbound Emerg Dis.* 2022 Sep;69(5):e1936-e1950. doi: 10.1111/tbed.14528. Epub 2022 Mar 30. PMID: 35306749; PMCID: PMC9790522.
11. **Gunasekera, U.C.**, Sivasothy, A., Wedasingha, N., Thayaparan, S., Rotewewa, B., Muralithas, M.Baumann, M.P.O., Punyapornwithaya, V., Analyzing the Foot and Mouth Disease outbreak as from 2008 to 2014 in cattle and buffaloes in Sri Lanka. *Preventive Veterinary Medicine* 148, 78–88, 2017.
12. **Umanga C. Gunasekera**, Susuji Wickramasinghe, Ganga Wijesinghe, R.V.J. Rajapakse, Gastrointestinal parasites of captive primates in the national zoological gardens of Sri Lanka. *Taprobanica, The journal of Asian Biodiversity*, ISSN 1800-427X, vol 4 (1), pp 37-40, 2012.
13. **Umanga C Gunasekera**, Mahalingam Muralithas, Maximilian Baumann, Food safety monitoring and surveillance system for Sri Lanka, *Sri Lanka veterinary journal*, vol 63(2B), pp 31-34, 2016.
14. Sivapiragasam Thayaparan, Robertson I D, Fairuz A, Suut L, **Gunasekera U. C**, Abdullah M T, (2015). Sero epidemiological study of leptospirosis among the communities living in peri urban areas of Sarawak, Malaysia. *The Medical Journal of Malaysia* vol 70 (5), pp 281-287, 2015.

Clinical Abstracts

1. Umanga C. Gunasekera, P.G.A. Pushpakumara and L.N.A. De Silva, Endotoxemia in Horses in Sri Lanka - Case Report. *Vet Scan Veterinary Journal*, ISSN 0973-6980, vol 7 (2), pp 73-81, 2013.
2. Umanga C. Gunasekera, J.L.C.S Perera, C. Dushyanthan, A.M.P. Abeyasinghe, G.D.R.K. Perera, A.A.A.W.K. Amarasinghe, P.G.A. Pushpakumara and L.N.A. De Silva. An outbreak of acute typhlocolitis in group of thoroughbred horses in single stable. *Sri Lanka Veterinary Association Scientific Sessions* pp 29, 2012, May.
3. W.A.S.M. Premarathne, K. Nizanantha, W.U.C. Gunasekera, A.M.P. Abeyasinghe, L.N.A. De Silva and P.A.B.D. Alexander. Uterine torsion in cattle and buffaloes; a case study. *Sri Lanka Veterinary Association Scientific Sessions* pp 4, 2013, June. J.M.R.V. Wijayarathne, W.M.D.S. Wanninayake, Y.H.P.S.N. Kumara, K. Nizanantha, W.U.C. Gunasekera and L.N.A. De Silva. Urea toxicities in a group of dry cows in an upcountry dairy farm. *Sri Lanka Veterinary Association Scientific Sessions* pp 7, 2013, June.
4. H. E. Abeygunawardena, I.D. Silva, D.D.N. De Silva, A. Dangolla, J.U. Weerasinghe, A.M.R. Weerasinghe, I.S. Hewapathirana, A.M.R. Bandara, W.U.C. Gunasekera and A.C. Karawita. An embolectomy in the left external iliac artery in a paraplegic dog, *Peradeniya University research sessions* 2010, November.

Academic services

1. Topic Editor *Journal of Frontiers in Ecology and Evolution*
2. Reviewer *Journal of Veterinary Preventive Medicine*
3. Reviewer *Journal of Transboundary and Emerging Diseases*
4. Reviewer *journal of BMC Infectious Diseases*
5. Volunteer graduate student travel grant reviewer for COGS (Council of Graduate Students), University of Minnesota 2018 and 2019.
6. CAHFS newsletter monthly topic on "Feasibility of global animal disease control strategies in

Asia", April 01, 2021.

7. Sri Lanka Veterinary Journal, letter to the editor, "Implication for a successful FMD control Program in Sri Lanka", December, 2021.
8. Newspaper article on Food poisoning and trace back system of food products, 9 March 2015, The daily newspaper Sri Lanka.
9. Provided epidemiological support by conducting discussions with the government veterinarians regarding identifying different risk zones for foot and mouth disease in Sri Lanka to optimize the FMD control program in the country May, 2024.

Professional memberships

American College of Veterinary Preventive Medicine
Sri Lanka Veterinary Association

Presentations

1. **Gunasekara, Umanga**, Alkhamis Moh A., Puvanendiran Sumathy, Das Moumita, Kumarawadu Pradeep L, Sultana Munawar, Hossain Anwar, Arzt Jonathan, Perez Andres. 2024, "Ecological Niche Modeling for Surveillance of Foot-and-Mouth Disease in South Asia". 17th International Symposium of Veterinary Epidemiology and Economics (ISVEE), November 8 - 15, 2024. Oral presentation.
2. **Gunasekara, Umanga**, Miranda R. Bertram, Nguyen Van Long, Phan Quang Minh, Vo Dinh Chuong, Andres Perez, Jonathan Arzt, and Kimberly VanderWaal. "Phylogeography as a Proxy for Population Connectivity for Spatial Modeling of Foot-and-Mouth Disease Outbreaks in Vietnam." 16th International Symposium of Veterinary Epidemiology and Economics (ISVEE), August 7 - 12, 2022. Oral presentation.
3. **Umanga Gunasekara**, Miranda Bertram, Do.H Dung, Nguyen Phuong, Vo.V Hung, Nguyen V Long, Minh Phan, Andres Perez, Jonathan Arzt, Kimberly VanderWaal, Using phylogeography as a proxy for population connectivity for spatial modeling of outbreak data of Foot and Mouth Disease in Vietnam. "Epidemics8 - 8th International Conference on Infectious Disease Dynamics", 2021 November. Oral presentation.
4. **Gunasekara, U.**, Bertram, M.R.; Dung, D.H.; Hoang, B.H.; Phuong, N.T.; Hung, V.V.; Long, N.V.; Minh, P.Q.; Vu, L.T.; Dong, P.V.; Perez, A.; VanderWaal, K.; Arzt, J. Use of Slaughterhouses as Sentinel Points for Genomic Surveillance of Foot-and-Mouth Disease Virus in Southern Vietnam. Global Foot and Mouth Disease Research Alliance (GFRA) Scientific sessions, 2021, November. **Invited presentation.**
5. **U. C. Gunasekera**, J. Arzt, R. Ranjan, J. Biswal, A. Perez, K. VanderWaal, Modeling the spatiotemporal dynamics of foot-and-mouth disease during mass vaccination programs in India. 15th International symposium of veterinary epidemiology and economics. 2018, November 12-16, Chiangmai, Thailand. Oral presentation.
6. **Umanga C. Gunasekera**, Veerasak Punyapornwithaya, Nihal Wedasingha, Maximilian P.O. Baumann, Marcus G. Doherr, Recent Foot and Mouth Disease outbreaks and risk factors identified in Sri Lanka, Joint AITVM-STVM conference, September 5, (2016), Berlin, Germany. Oral presentation.
7. **Gunasekera Umanga C**, Punyapornwithaya Veerasak, Wedasingha Nihal, Chaisowwong Warangkhan, Baumann Maximilian P.O., Doherr Marcus G. Identification of Risk Factors for the Recent Foot and Mouth Disease Outbreak in Sri Lanka, Global Foot and Mouth Disease Research Alliance (GFRA)(20-22, October, 2015, Hanoi, Vietnam. Oral presentation.

8. **Umanga C Gunasekera**, Veerasak Punyapornwithaya , Nihal Wedasingha, Maximilian P.O. Baumann Marcus G. Doherr, First result from a case control study for the identification of risk factors associated with Foot and Mouth disease in Anuradhapura district of Sri Lanka. The 4th Food safety and zoonoses symposium for Asia pacific, Oral presentation, Chiangmai, Thailand, (2015), August.
9. **Umanga C. Gunasekera**, J.L.C.S Perera, C. Dushyanthan, A.M.P. Abeysinghe, G.D.R.K. Perera, A.A.A.W.K. Amarasinghe, P.G.A. Pushpakumara and L.N.A. De Silva. An outbreak of acute typhlocolitis in a group of thoroughbred horses in a single stable. Sri Lanka Veterinary Association Scientific Sessions (2012), May. Oral Presentation.
10. **U. C. Gunasekera**, K. A. B. T. Bandara, S. Wickramasinghe, G. Wijesinghe and R. P. V. J. Rajapakse. A preliminary study on gastrointestinal parasites of captive primates in the national zoological gardens, Dehiwala, Sri Lanka. Peradeniya University research sessions 2011, November. Oral Presentation.
11. **U. Gunasekera**, M.R. Bertram, Le T. Vu, Do H. Dung, Bui H. Hoang, Nguyen T. Phuong, Vo V. Hung, Long Nguyen, A. Perez, J. Arzt, K. VanderWaal, EuFMD virtual conference, 2020. Poster presentation of "Molecular surveillance of foot-and-mouth disease virus at slaughterhouses in Vietnam".
12. **U. Gunasekera**, M.R. Bertram, Le T. Vu, Do H. Dung, Bui H. Hoang, Nguyen T. Phuong, Vo V. Hung, Long Nguyen, A. Perez, J. Arzt, K. VanderWaal. Poster presentation of "Molecular surveillance of foot-and-mouth disease virus at slaughterhouses in Vietnam", CRAWAD Animal health virtual conference, 2020.
13. **U. C. Gunasekera**, J. Arzt, R. Ranjan, J. Biswal, A. Perez, K. VanderWaal, Poster presentation, "Modeling the spatiotemporal dynamics of foot-and-mouth disease during mass vaccination programs in India" in the 3rd international conference of one medicine one science (ICOMOS), Minneapolis, USA, 2018, April. Poster presentation.

Teaching experience

1. Contributing to Progress Vet South East Asia E-Book 2 chapter, Data Analysis Applied to ASF Control, module "Understanding factors that affect disease", 2023. Reviewing Progress Vet E-Books and resource catalog of free online resources to use for content creation. Conducting webinar sessions and grading course work of the participants.
2. Completed the Designing and Delivering Online Learning Program conducted by the Center for Educational Innovations, University of Minnesota, Fall 2024.
3. Coordinating the 'Ecological Niche Modeling' workshop conducted by Moh Alkhamis at the Center for Animal Health and Food Safety, March 23, 2023.
4. Completed the Teaching Assistant Professional Development (TAPD) Program offered through the Center for Educational Innovation (CEI), University of Minnesota May 13, 2021.
5. Completed CVM instructor development course, College of Veterinary Medicine, University of Minnesota, July 28, 2021.
6. Zoom presentation about the topic "Animal Disease Surveillance" to the MVSC students (batch 2020), Faculty of Veterinary Medicine, University of Peradeniya, Sri Lanka, March 11, 2021.
7. Zoom presentation on "Bayesian Disease Mapping" for Chiangmai University researchers, VPHACP, Thailand on January 23, 2021.

8. Probationary Lecturer 5/02/2011–30/09/2013

Faculty of Veterinary Medicine, University of Peradeniya, Peradeniya (Sri Lanka)

- Conducted demonstrations for final-year veterinary students on clinical cases of large animal health.
- Participate in research work conducted by the Department of Health Screening farm animal health.
- Attended clinical cases (equine, bovine, and poultry) both at the hospital and by ambulatory visits.
- Conducted field visits regarding Poultry farms and Dairy farms.
- Provided knowledge and advice to farmers to uplift their livelihood by avoiding health and zoonotic problems, selecting good animals for rearing purposes, and preventing the spreading of contagious diseases and zoonotic infectious diseases.

9. Moumita Das, Mentoring graduate student on her work on spatial analysis of rabies outbreaks in Kenya. University of Minnesota. 2024.

Conferences

1. Participated in the Allen D. Leman Swine Conference organized by the College of Veterinary Medicine, University of Minnesota. September 2022.
2. Participated in the Bayesian Network Workshop conducted by Charles Sturt University, Australia September 7th, 2021.
3. Participated in the workshop on BEAST analysis, at the University of Washington Bio statistic summer camp, Washington, Seattle, 2019.
4. Participated in the workshop "Introduction to Bayesian Disease Mapping and Advanced Bayesian Disease Mapping", Medical University of South Carolina, 2018.
5. Participated in the 3rd SAARC FMD Regional Roadmap meeting workshop, Colombo, Sri Lanka, 2016.
6. Participated in the workshop ICI3D Clinic on Dynamical Approaches to Infectious Disease Data conducted by the Centre of Excellence in Epidemiological Modelling and Analysis (SACEMA).
7. Participated in the 3rd international congress on pathogens at the Human-Animal Interface (ICOPHA): One health for sustainable development, Chiangmai, Thailand, 2015.
8. Participated in the 4th Food Safety and Zoonoses Symposium for Asia Pacific and the 2nd regional Eco Health symposium, Chiangmai, Thailand, 2015 August.
9. Participated in the Risk Communication Workshop conducted by Professor Will Huston in the Veterinary Public Health Center for Asia Pacific in 2014, Chiangmai University.
10. Participated in the workshop on orthopedic surgery conducted by Prof Richard Read from Murdoch University, Australia in Sri Lanka.
11. Participated in the short course on the Health and Tranquilization of Captive Elephants conducted by Post Graduate Institute of Science, University of Peradeniya in 2009.
12. Participated in a Workshop on Symposium of Water Resources Management and Landscape Management in 2009 conducted by PGIS, University of Peradeniya.

Awards

1. Scholarship holder of German Academic Exchange Service (Deutscher Akademischer Austausch Dienst-DAAD) Scholarship to complete the Master's degree in Veterinary Public Health. €25000.
2. International travel grant from the University of Minnesota Global Programs and Strategy Alliance to participate in 17th International Symposium on Veterinary Epidemiology and Economics, 11-15 November 2024, Sydney, Australia for oral presentations "FMD dynamics in Vietnam" and "Ecological niche modelling to detect FMD risk in South Asia". \$1400
3. Travel award to participate in Global Foot and Mouth Disease Research Alliance (GFRA) 20-22, October, 2015, Hanoi, Vietnam from GFRA for oral presentation "Identification of Risk Factors for the Recent Foot and Mouth Disease Outbreak in Sri Lanka".

4. DAAD Travel Award to participate in the AITVM-SVTM Conference in Berlin, Germany, 2016 including the airfare for oral presentation of “Recent Foot and Mouth Disease outbreaks and risk factors identified in Sri Lanka”.
5. College of Veterinary Medicine, University of Minnesota, Travel award to participate in ICI3D Clinic on Dynamical Approaches to Infectious Disease Data, Florida. Organized by the DSI-NRF Centre of Excellence in Epidemiological Modelling and Analysis (SACEMA) (2017, December). \$1000.
6. OGP Professional Development Award, University of Minnesota (2018) to participate in the workshop on Introduction to Bayesian Disease Mapping and Advanced Bayesian Disease Mapping. University of South Carolina.\$1400
7. Award for the best presentation at the Sri Lanka Veterinary Association Scientific Sessions (2012). Umanga C. Gunasekera. An outbreak of acute typhlocolitis in a group of thoroughbred horses in single stable. Sri Lanka Veterinary Association Scientific Sessions (2012), May. Rs1000.
8. Award for the best presentation at the Sri Lanka Veterinary Association Scientific Sessions (2013) Umanga Gunasekera. Urea toxicities in a group of dry cows in an upcountry dairy farm. Sri Lanka Veterinary Association Scientific Sessions, (2013), June. Rs1000.